05 · Case 11, question 5: the hormone tick table

The words of this track, as it says them. Bold brown is the examiner.

Case 11, question five. The hormone table. Five marks.

Complete the table about the roles of hormones in breast development, milk production and release, by ticking progesterone, oestrogen, prolactin or oxytocin for each row. Row one: stimulates growth of milk ducts and glands. Row two: increases blood flow to the breasts. Row three: stimulates development of the lobules and alveoli. Row four: stimulates milk production and the development of mammary glands. Row five: stimulates milk ejection, by making the myoepithelial cells around the milk ducts contract.

Five rows, five marks, and one picture does all of it.

The hook: oestrogen builds the pipes. Progesterone builds the factories. Prolactin fills them. Oxytocin opens the taps.

Row one, growth of the milk ducts and glands. Tick oestrogen and progesterone, both. Her deck gives the ducts to oestrogen, from puberty, and her lactation case quiz says the ducts and alveoli develop under the influence of progesterone. So this row takes two ticks.

Row two, blood flow to the breasts. Tick oestrogen. Be aware: none of her files answers this row, so this tick is the podcast's own reading. It's the best fit, but it isn't her key.

Row three, the lobules and alveoli: progesterone, in pregnancy. The factories.

Row four, milk production and the development of the mammary glands: prolactin, during lactation. It fills the factories.

Row five, milk ejection, the myoepithelial cells squeezing: oxytocin. The let-down, at feeding time. It opens the taps.

The traps. Progesterone does the lobules and alveoli; oestrogen does the ducts. Second, the same swap as last time: prolactin produces, oxytocin ejects. And if a row turns up that you've never seen answered, like blood flow, don't panic. Oestrogen.

Once more. The five rows.

Ducts and glands grow: oestrogen and progesterone. Blood flow to the breasts: oestrogen. Lobules and alveoli: progesterone. Milk production: prolactin. Milk ejection, the myoepithelial cells contracting: oxytocin. Pipes, factories, fill, taps.